

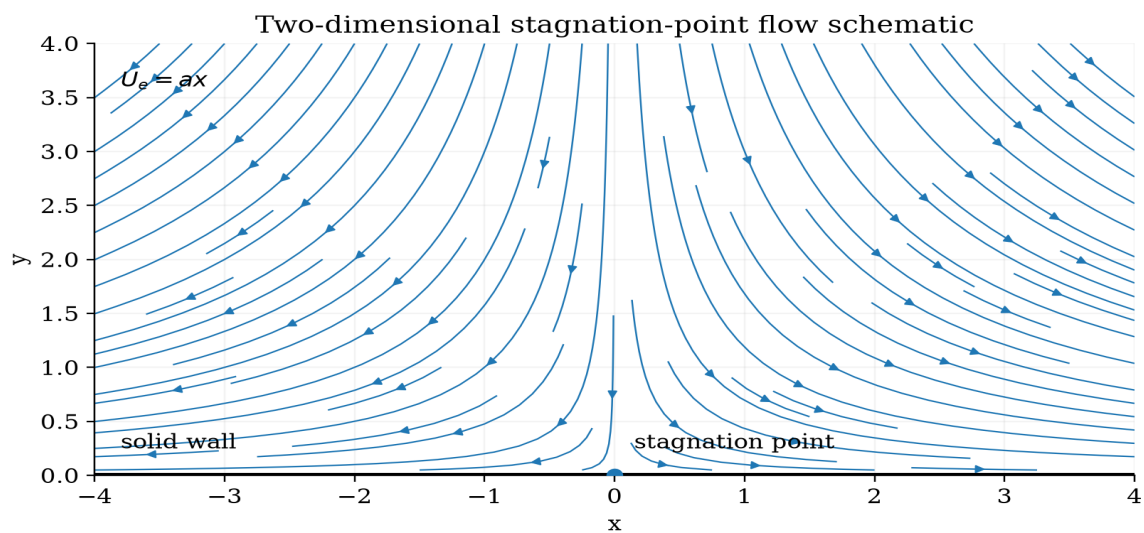
NUMERICAL INVESTIGATION OF STAGNATION-POINT BOUNDARY-LAYER FLOW

Viscous Flow - Assignment 01

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A numerical study comparing direct boundary-value collocation, Newton shooting with adaptive IVP integration, and fixed-step RK4 shooting with a propagated sensitivity equation.

1. Objective

The objective of this assignment is to obtain the similarity solution of the two-dimensional stagnation-point boundary layer and to examine the consistency of several numerical solution strategies. The problem is a nonlinear third-order boundary-value problem in the similarity coordinate η .

Governing similarity equation

$$f''' + f f'' - (f')^2 + 1 = 0$$

Boundary conditions

$$\begin{aligned} f(0) &= 0 \text{ (impermeable wall)} \\ f'(0) &= 0 \text{ (no-slip condition)} \\ f'(\infty) &= 1 \text{ (matching with the outer flow)} \end{aligned}$$

For numerical work, the semi-infinite domain is truncated at $\eta_{\infty} = 7$. The main quantity extracted from the solution is the dimensionless wall-shear parameter $f''(0)$.

2. First-Order Form

Introducing $y_1 = f$, $y_2 = f'$ and $y_3 = f''$ converts the third-order equation to the system below.

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= y_3 \\ y_3' &= y_2^2 - y_1 y_3 - 1 \end{aligned}$$

3. Numerical Methods

3.1 Direct BVP collocation

The complete boundary-value problem is solved directly with a collocation-based BVP solver. A smooth tanh-type velocity estimate is used as the initial profile, after which the solver refines the mesh where the solution requires additional resolution.

3.2 IVP shooting and Newton-Raphson correction

The unknown initial value $s = f''(0)$ is treated as a shooting parameter. For a trial s , the equations are integrated from $\eta = 0$ to $\eta = 7$ and the terminal residual is defined as

$$R(s) = f'(7; s) - 1$$

Newton's method then updates the shooting parameter using $s(\text{next}) = s - R(s)/R'(s)$. The derivative is evaluated using a small forward perturbation in the shooting parameter.

3.3 Fixed-step RK4 with sensitivity

A classical fourth-order Runge-Kutta integrator is implemented explicitly. In addition to the physical variables, the sensitivities of f , f' and f'' with respect to the shooting parameter are integrated simultaneously. This supplies the Newton derivative from the same numerical trajectory and provides an independent implementation of the shooting method.

4. Boundary-Layer Quantities

The converged velocity profile provides several standard dimensionless measures. The displacement thickness, momentum thickness and shape factor are evaluated as

$$\begin{aligned}\delta^* &= \int_0^{\eta_{\infty}} (1 - f') \, d\eta \\ \theta &= \int_0^{\eta_{\infty}} f'(1 - f') \, d\eta \\ H &= \delta^* / \theta\end{aligned}$$

The 99 percent boundary-layer thickness δ_{99} is obtained by linear interpolation at the point where $f'(\eta) = 0.99$.

5. Numerical Results

Method	$f''(0)$	Abs. error vs ref.
Direct BVP	1.232587656759	3.24e-09
IVP + Newton	1.232587656723	3.28e-09
RK4 + sensitivity	1.232587656595	3.41e-09
Reference	1.232587660000	-

Quantity	Value
δ_{99}	2.379462
Displacement thickness δ^*	0.647917
Momentum thickness θ	0.292328
Shape factor H	2.216406

The three approaches give nearly indistinguishable similarity profiles.

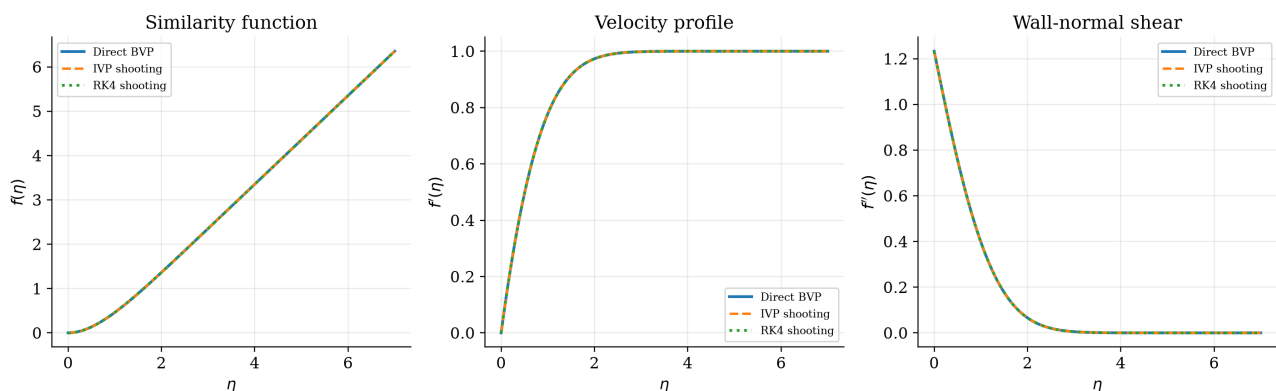


Figure 1. Similarity function, velocity and shear profiles.

6. Convergence and Sensitivity Studies

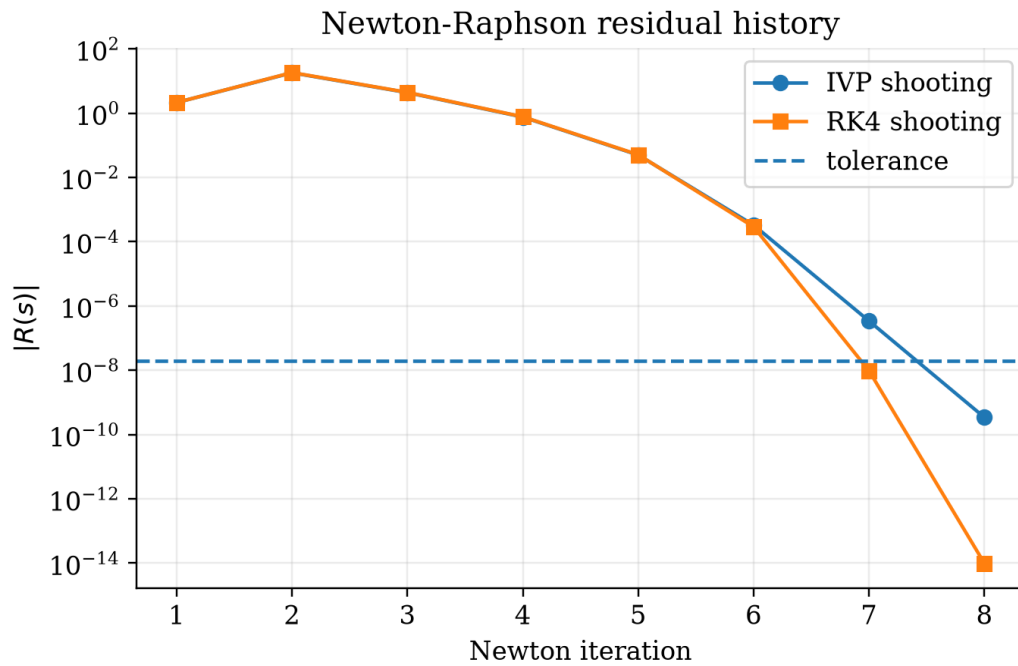


Figure 2. Newton-Raphson residual reduction for the two shooting formulations.

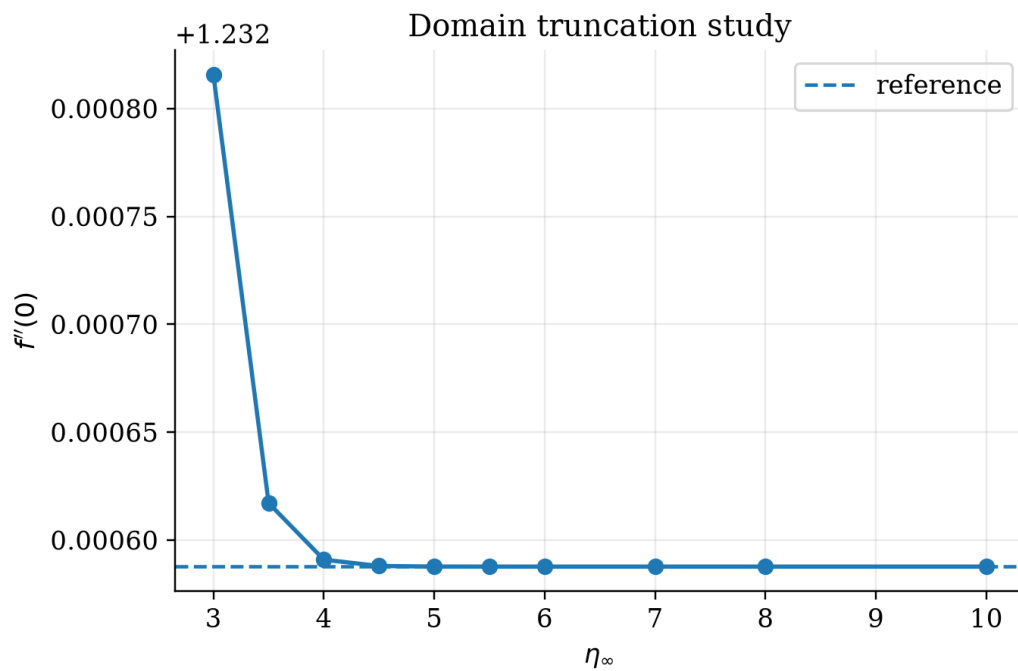


Figure 3. Effect of the finite computational endpoint η_∞ .

7. RK4 Refinement and Derived Quantities

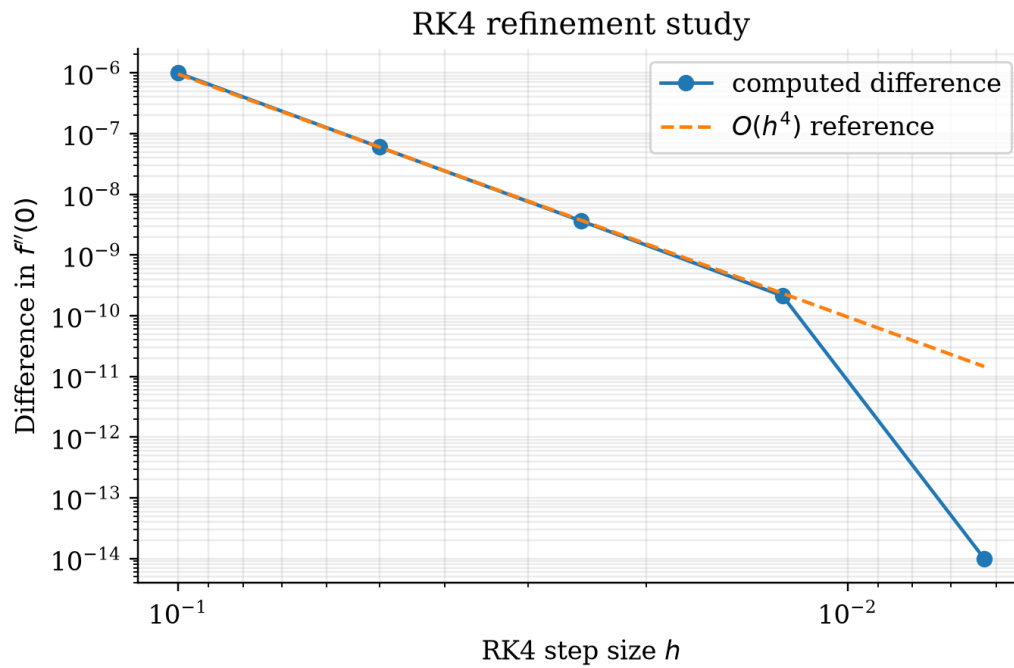


Figure 4. Fixed-step RK4 refinement study with an $O(h^4)$ reference trend.

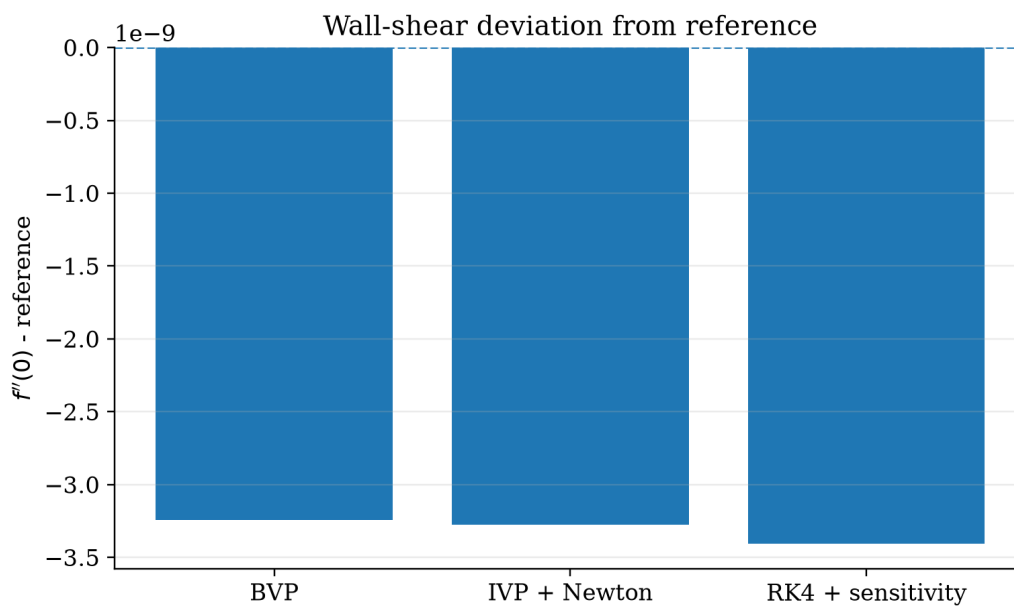


Figure 5. Deviation of the computed wall shear from the reference value.

8. Boundary-Layer Measures and Flow Configuration

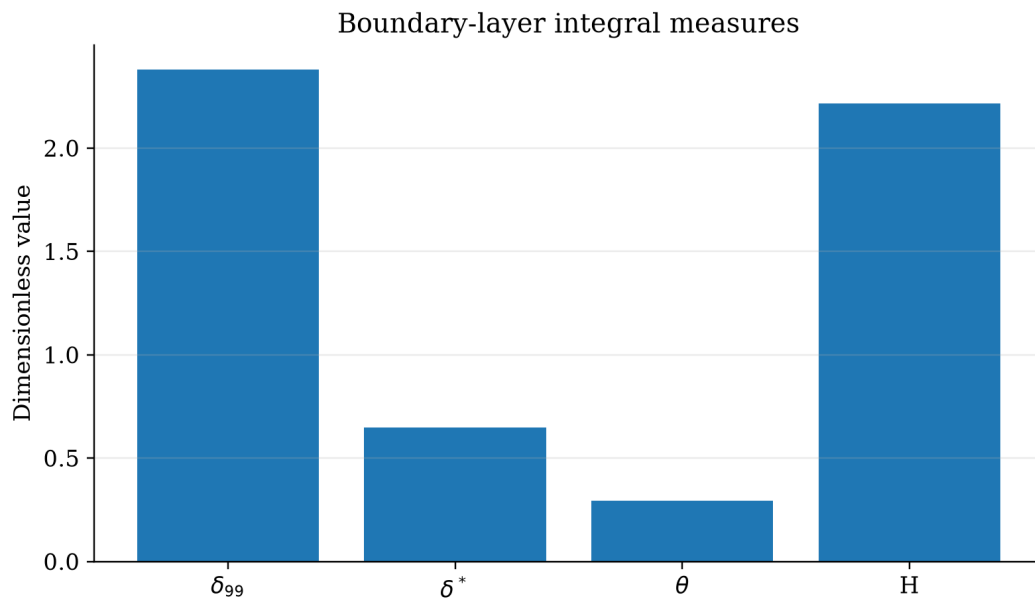


Figure 6. Dimensionless boundary-layer measures extracted from the RK4 solution.

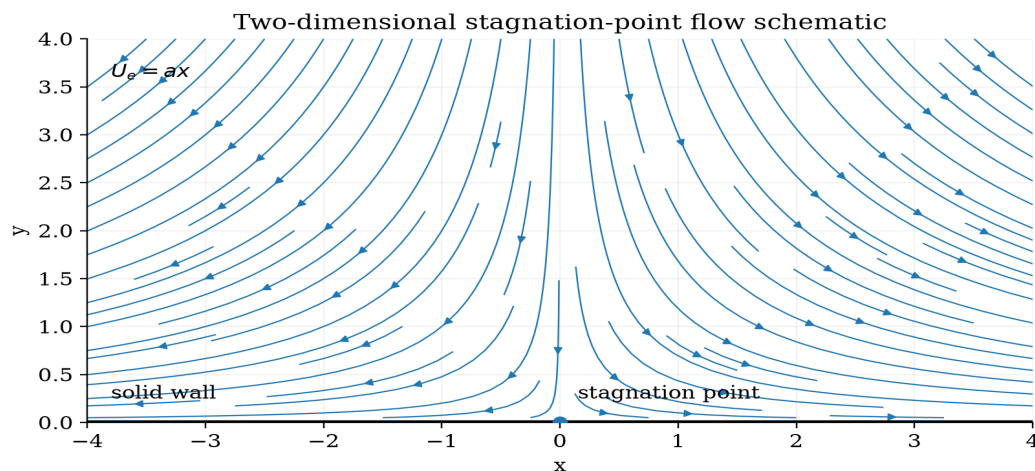


Figure 7. Two-dimensional stagnation-point flow schematic used to interpret the similarity solution.

9. Discussion

The three numerical formulations converge to essentially the same wall-shear parameter, providing a cross-check between direct and shooting formulations. Direct BVP collocation solves the boundary conditions simultaneously, whereas shooting converts the unknown wall shear into a scalar root-finding problem. The sensitivity-based RK4 implementation is particularly transparent because the Newton derivative is propagated as part of the state vector rather than reconstructed from separate solves.

The computational-domain study shows the influence of replacing infinity with a finite endpoint. Once the endpoint is sufficiently far into the outer-flow region, the wall-shear estimate changes very little. The RK4 refinement study provides an additional consistency check on the fixed-step implementation and its expected fourth-order behavior.

10. Conclusion

The stagnation-point similarity problem was solved using direct collocation, adaptive IVP shooting with Newton correction, and a fixed-step RK4 sensitivity formulation. The calculated wall-shear parameter is 1.23258766 to the displayed precision, and the associated boundary-layer integral quantities are reproduced consistently. The repository contains the numerical source, figures, tabulated results, and this report so that the complete calculation can be reproduced.

References

- H. Schlichting and K. Gersten, Boundary-Layer Theory, Springer.
- F. M. White, Viscous Fluid Flow, McGraw-Hill.
- V. M. Falkner and S. W. Skan, Some approximate solutions of the boundary layer equations, Philosophical Magazine.